

THATO BARUTI

+44 07759 037559 | tbofficial0508@gmail.com | github.com/ThatoBaruti619 | linkedin.com/in/thato-baruti
ML · Optimisation and Uncertainty · Simulation Pipelines · Engineering AI Platforms

Applied

PROFILE

Applied ML engineer completing an MSc in Computational Fluid Dynamics at Cranfield University, building deployment-facing AI systems for simulation-driven engineering and CAE workflows. My work spans dataset and graph construction, model training, evaluation gates, uncertainty calibration, optimisation workflows, higher-fidelity CFD validation, and API/UI layers that expose models safely. I am strongest at turning research ideas into auditable engineering tools with clear contracts, traceability, and realistic limitations rather than treating model accuracy in isolation as the finish line.

SELECTED IMPACT

- **Built an end-to-end engineering AI stack:** from dataset preparation and graph building through model training, calibration, report generation, and deployment-facing FastAPI and Gradio interfaces for wing-design recommendation.
- **Made optimisation and validation operational:** implemented requirement-conditioned Pareto search over 63,393 candidate designs, extracted 1,012 non-dominated candidates, and connected the optimisation layer to auditable higher-fidelity CFD validation queues.
- **Established evidence-driven model release gates:** created baselines, strict compare tools, locked-model audits, failure-slice diagnostics, and reliability ranking so model promotion depends on measured behaviour across regimes rather than headline averages alone.

ENGINEERING EXPERIENCE

AeroGenic Applied ML and Platform Stack

2025 – Ongoing

MSc Research, Cranfield University

- Built the AeroGenic codebase as a working engineering AI stack around aerodynamic coefficient prediction, combining graph neural surrogates, metadata baselines, hybrid models, calibration tools, report generation, and deployment-facing model-selection logic.
- Added FastAPI endpoints and a Gradio workbench that convert uploaded geometry and flight requirements into traceable wing-design outputs, exported geometry, confidence scores, source-case provenance, and machine-readable JSON reports.
- Delivered strong held-out model results while keeping deployment claims strict: the MeshGraphNet-style model reached 3.18% for C_L , 2.60% for C_D , and 3.83% for C_{My} ; the polar model improved C_D to 2.49%; the target-wise champion ensemble achieved 2.87% deployment-weighted held-out MAPE.

Simulation, Optimisation, and Validation Orchestration

2025 – Ongoing

Cranfield University

- Implemented requirement-conditioned Pareto optimisation across drag, trim, lift overshoot, span, and training-manifold risk objectives; generated 63,393 candidates, retained 15,031 feasible lift-band designs, and extracted 1,012 non-dominated solutions.
- Built candidate-selection logic for higher-fidelity CFD follow-up using residual size, uncertainty width, aerodynamic stress variables, airfoil coverage, and representative anchors so validation effort is targeted rather than ad hoc.
- Connected optimisation outputs to design-traceability artefacts, allowing each recommended geometry to carry its model source, confidence layer, nearest failure slice, and validation status into the next engineering gate.

HPC CFD Commissioning and Multi-Fidelity Correction

2025 – Ongoing

Cranfield University HPC Infrastructure

- Automated OpenFOAM case generation and debugging across meshing, decomposition, solver stability, module-path propagation, and function-object compatibility issues; encoded the structural fixes directly into the generator so regenerated cases inherit them automatically.
- Managed a 32-case RANS validation campaign with 29 returned cases and 26 strict-pass evidence cases, turning the CFD layer into an operational validation gate rather than a one-off visual check.
- Used the higher-fidelity CFD evidence to build grouped geometry-held-out residual correction layers, reducing corrected MAE to 0.0677 for C_L , 0.0104 for C_D , and 0.401 for C_{My} .

Workflow Automation and Technical Delivery

2024 – Ongoing

Cranfield University

- Scripted multi-stage simulation and ML workflows in Python and C++ across OpenFOAM, ANSYS Fluent, and UCNS3D, covering geometry preparation, meshing, solver execution, extraction, training, evaluation, and figure generation.

- Built tools for metrics, baselines, model comparison, training-log visualisation, report figures, and publication-ready summaries so new checkpoints are released against a consistent evidence standard.
- Worked across the product boundary between model development and interface design, including geometry-file ingestion and export routes for STL, OBJ, and IGES-style workflows.

EDUCATION

MSc Computational Fluid Dynamics
Cranfield University, UK
On track for distinction; selected modules include Data Analysis & Post-Processing (87%), Introduction to Fluid Mechanics & Heat Transfer (87%), Turbulence Modelling, Numerical Methods & HPC, Grid Generation, and Numerical Modelling.

Oct 2025 – Aug 2026 (Expected)

BEng Aerospace Engineering
University of Sheffield, UK
Awards: University of Sheffield International Undergraduate Scholarship; Botswana Government Top Achievers Scholarship.

Jun 2025

ADDITIONAL ENGINEERING EXPERIENCE

AeroMAP UAS Build
Performance Lead, University of Sheffield

Oct 2023 – May 2024

- Directed a four-person technical team, owned simulation-to-validation workflow across five components, and delivered +25% stability and +15% endurance validated against flight test.

SunStratos High-Altitude Surveillance Glider
Structures Lead, University of Sheffield

Feb 2023 – Jun 2024

- Led a 12-member cross-functional team from design through to manufactured prototype and improved delivery speed by 33% through structured ownership and clear task allocation.

TECHNICAL SKILLS

Languages and ML	Python; C++; PyTorch; PyTorch Geometric; NumPy; scikit-learn; FastAPI; Gradio
ML Engineering	Graph neural networks; surrogate modelling; calibration; reliability ranking; residual correction; optimisation pipelines; model comparison gates
Platform	REST-style APIs; JSON reporting; workflow traceability; figure and report generation; deployment-facing interface design
Simulation	OpenFOAM; ANSYS Fluent; UCNS3D; Linux HPC workflows; PBS; SLURM; MPI-parallel job pipelines; ParaView; Tecplot
Domain	Aerodynamics; wing design; CFD validation; simulation-driven product and CAE workflows